

Review

Advances in Object Detection and Localization Techniques for Fruit Harvesting Robots

Xiaojie Shi, Shaowei Wang, Bo Zhang, Xinbing Ding, Peng Qi , Huixing Qu, Ning Li, Jie Wu and Huawei Yang * 

Shandong Academy of Agricultural Machinery Sciences, Jinan 250100, China; itismyway163@163.com (S.W.); xinb.ding@gmail.com (X.D.); qi-peng@139.com (P.Q.); palm06@163.com (N.L.); cangzhibiwujie@163.com (J.W.)

* Correspondence: yhw105@163.com

Abstract: Due to the short time, high labor intensity and high workload of fruit and vegetable harvesting, robotic harvesting instead of manual operations is the future. The accuracy of object detection and location is directly related to the picking efficiency, quality and speed of fruit-harvesting robots. Because of its low recognition accuracy, slow recognition speed and poor localization accuracy, the traditional algorithm cannot meet the requirements of automatic-harvesting robots. The increasingly evolving and powerful deep learning technology can effectively solve the above problems and has been widely used in the last few years. This work systematically summarizes and analyzes about 120 related literatures on the object detection and three-dimensional positioning algorithms of harvesting robots over the last 10 years, and reviews several significant methods. The difficulties and challenges faced by current fruit detection and localization algorithms are proposed from the aspects of the lack of large-scale high-quality datasets, the high complexity of the agricultural environment, etc. In response to the above challenges, corresponding solutions and future development trends are constructively proposed. Future research and technological development should first solve these current challenges using weakly supervised learning, efficient and lightweight model construction, multisensor fusion and so on.

Keywords: object detection; object localization; deep learning; computer vision; fruit-harvesting robots



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1. Introduction

With the deterioration of climatic conditions in recent years, many regions of the world are facing extreme weather events such as droughts, high temperatures, floods and wildfires [1,2]. In addition, the world's population is increasing and the area of arable land is decreasing. This requires the production of more grain, vegetables and fruits in the agricultural system to meet the needs of the people [3,4]. Fruit-harvesting work still requires a large number of people to perform it manually. In order to reduce costs and improve labor efficiency, many intelligent vegetable- and fruit-harvesting robots have been developed [5–8]. The fruit- and vegetable-harvesting environment is complex and susceptible to rain, snow, light and other uncertain factors, which brings greater difficulties to automated-harvesting operations. The efficient fruit- and vegetable-harvesting robot requires accurate object recognition and three-dimensional positioning support, because the accuracy of object detection and positioning directly affects the automated-harvesting efficiency, the degree of crop harvest loss and whether the harvesting robot itself is damaged by collision [9].

Since the 1990s, the harvesting robot has been rapidly developed and a large number of new methods have emerged in the field of object recognition. In the early days, researchers used traditional digital image processing technology to achieve object detection and recognition based on color (RGB, HSV, Lab, HSI), texture, shape and other features extraction [10,11]. Thanks to advances in machine learning, some detection algorithms based on K-means clustering, K-nearest neighbor clustering, Bayes Classifier, Haar-like features and the AdaBoost algorithm, and the support vector machine (SVM) algorithm have been proposed [12–18]. Deep learning, which originated from convolutional neural networks (CNNs), can acquire more abstract categories or features of high-level attributes [13,19]. In terms of fruit detection, some people have developed novel algorithms for an automated vegetable- and fruit-harvesting robot based on Alex Krizhevsky networks (AlexNets), visual geometry group networks (VGGNets), regions with convolutional neural networks (R-CNNs), faster region-convolutional neural networks (Fast R-CNNs), faster region-convolutional neural networks (Faster R-CNNs), the single-shot multibox detector (SSD), You Only Look Once (YOLO) and so on [20–27].

The representative research results of object recognition and localization of picking robots in recent years are introduced in this paper. Kapach et al. briefly described the vision system of automatic fruit-harvesting robots, and comprehensively summarized the strengths and weaknesses of various methods in current research areas. Most of the research works involved the color of the object, but little research works were carried out on the detection and recognition of objects with inconspicuous color such as bagged fruits [28]. Fu et al. summarized the application of various RGB-D cameras and related algorithms for fruit detection and localization, pointing out the challenges faced by vision systems based on RGB-D cameras as well as future directions [29]. Wang et al. summarized the application of a detection algorithm based on CNNs in fresh fruits, compared different detection methods in different production stages and found that it possessed good application prospects [30]. Dhanush et al. provided a systematic summary of a vision system and related algorithms used in the production and inspection of agricultural products, detailing the role played by smart agriculture in agricultural production [31].

This paper focuses on the study of fruit detection and localization algorithms in harvesting robots based on DL methods. Firstly, the papers were retrieved from Google Scholar, ScienceDirect, Web of Science and other databases for the last decade using a combination of fruit recognition, fruit detection, fruit localization, harvesting robots and computer vision. After that, the above papers were evaluated and filtered to match the topic. Finally, the papers were classified and analyzed based on object detection and object localization. In this paper, the problems and challenges of fruit recognition are analyzed, and the future development trend is predicted, in order to provide some suggestions for future research on fruit detection and localization.

2. Fruit Detection Method

Compared with traditional algorithms, fruit detection methods based on deep learning have obvious advantages in object classification, detection and recognition. They utilize low-level features to form high-level features for the final tasks by introducing the multilayer perceptron structure. Since fruits in natural scenes have unstructured features such as random spatial distribution, presence of overlapping occlusion and diverse shapes, deep convolutional network methods can automatically learn features from training data [32]. The general steps of fruit detection are shown in Figure 1.

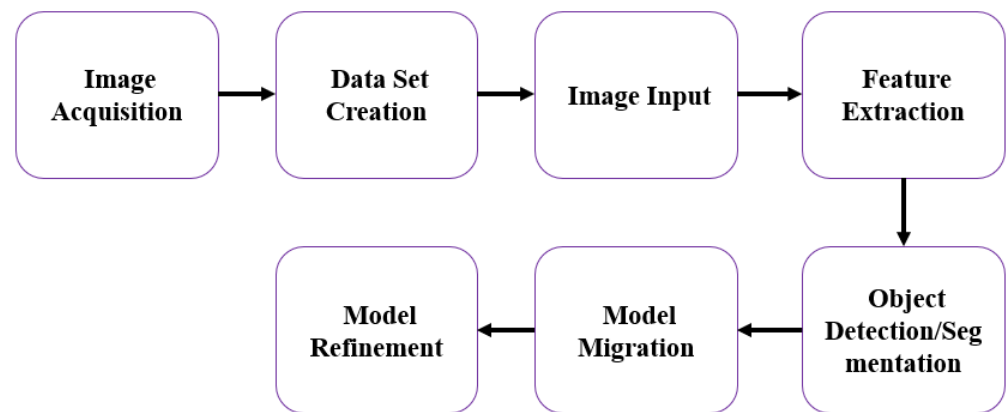


Figure 1. General steps of fruit recognition.

2.1. Dataset Preparation Method

Datasets act as a decisive role in model training and evaluation. They help DL models gain more powerful capabilities in object recognition, classification and segmentation tasks by training with diverse and widely covered image datasets. The first step is to collect data through different devices. The fruit images collected in the field are processed and analyzed in order to achieve the task of fruit detection in different scenarios, different varieties and different growth stages. In most cases, RGB cameras are used to obtain the color, texture, shape and other features of fruits, which is a more effective and less costly data acquisition tool. Zhao et al. used Canon cameras and Motorola smartphones to take pictures of grapes in Brazilian vineyards [33]. However, RGB cameras can only obtain the color information of the object, and cannot directly obtain the depth information of the object and the camera. To solve these problems, a number of RGB-D cameras are used to acquire color and depth images [29]. Kaukab et al. used an Intel RealSense D455 camera to collect images in high-density apple orchards under a number of different light conditions and angles [34]. However, most depth cameras can only acquire data up to 3.5 m, and longer distance data needs to be acquired through a robot such as light detection and ranging (LiDAR) scanners. Data preprocessing is performed for the features of different object recognition tasks, and the process includes both image augmentation, image labeling and so on. Image augmentation techniques are used to adjust the color, brightness, contrast etc. of the image, or to rotate, flip, crop etc. the image, so that the image input to the network is more suitable for a specific object recognition task or to achieve the expansion of the size of the dataset. Recently, some image enhancement based on deep learning such as adversarial training, GAN, CycleGAN and CVAE-GAN has also been proposed [35–38]. These algorithms usually contain the generator network and the discriminator network. The generator network is able to generate realistic enhanced images, while the discriminator network is able to distinguish the enhanced image from the original image. Through adversarial training of generators and discriminators, GANs can generate realistic images, repair low-quality images, expand datasets, perform style transfer and finally achieve the purpose of image enhancement. Image labeling, including object localization and category information, helps the model to accurately identify and locate various objects in the image.

Object recognition tasks based on deep learning require a certain scale of datasets for training. At present, most fruit detection methods based on deep learning are based on supervised learning, which trains the model by inputting a certain number of images and corresponding label files into the network to extract the features of the object and realize the task of object recognition [39]. Supervised learning is a common method to solve classification and regression problems. For the field fruit detection task, the training data determines the effect of the object detection task to a certain extent, and its scale depends

on the visual complexity of the fruit image and the type of deep learning model. The data-labeling process is time-consuming and inefficient, and for some specific recognition tasks, the data-labeling process requires guidance from relevant experts, so it is a meaningful problem to achieve faster and more accurate labeling of data samples and efficient training with small amounts of data in supervised learning. The unsupervised learning model only uses images as training data. It can avoid the above problems by learning the structure of the data, extracting distinguishable information or features from the data and mapping the input to a specific output, which is often used in the field of image segmentation [40].

2.2. Fruit Detection Method Based on DL

The historical evolution of different object detection algorithms based on DL is shown in Figure 2. The object detection method based on DL can be divided into the two-stage method (The algorithms highlighted in red in Figure 2) and the single-stage method. In the two-stage detection method, the region proposals are generated, and then they are precisely classified and located by a CNN. The typical methods are R-CNN, Fast R-CNN, Faster R-CNN and Mask R-CNN [41–44]. The single-stage object detection method eliminates the process of generating region proposals and directly generates the class probability and position coordinates of the object, which is simpler than two-stage object detection. The representative algorithms of single-stage object detection are SSD series, YOLO series and RT-DERT [45–47]. The related studies on the application of DL-based detection models in fruit harvesting are summarized in Table 1. Overall, two-stage detection methods have higher detection accuracy and one-stage detection methods have faster real-time detection. For some small fruits like apples, tomatoes, etc., which need to be detected in larger quantities and require high detection speeds, it is better to use single-stage detection algorithms for robotic harvesting. For some fruits that are heavily shaded and require high detection accuracy such as coconuts, it is better to use two-stage detection algorithms such as the Faster R-CNN.



Figure 2. Deep learning-based object detection algorithm, over time.

Table 1. Summary of related studies on application of DL-based detection models in fruit harvesting.

Crop Applied	Basic Model	Dataset	Detection Rate (%)	Inference Speed (ms/image)	Reference
Apple	YOLOv3	878 RGB images	92.2	19@3008 × 4512 px	[27]
	Faster R-CNN	800 RGB images	89.3	181@1920 × 1080 px	[48]
	mask R-CNN	1020 RGB images	97.31	-	[49]
	YOLOv5	1000 RGB-D images	92.4	17.2@1280 × 720 px	[50]
Tomato	Mask R-CNN	123 images	95	-	[51]
	SSD	1730 RGB-D images	98.85	-	[52]
	SSD	23,021 RGB images	85.31	24.75@300 × 300 px	[53]
	YOLOv5	2034 RGB images	87.3	-	[54]
Coconut	Faster R-CNN	2000 RGB images	89.4	3.124@516 × 516 px	[44]
	Faster R-CNN	900 RGB images	88.6	774@512 × 512 px	[55]
Mango	SSD	1730 RGB images	92.43	253@256 × 256 px	[26]
	SSD	572 RGB images	92.0	28.6@400 × 400 px	[56]
Oil palm	YOLOv4	490 RGB images	95	47.6@1920 × 1080 px	[57]
	YOLOv4	6719 RGB images.	92.6	12.11@1920 × 1080 px	[58]
Pineapple	YOLOv4	5,000,000 RGB images	97	14.3@800 × 600 px	[59]
Dragon Fruit	YOLOv7	2458 RGB images	84.4	-	[60]
Pomegranate	Mask R-CNN	400 RGB images	95.7	36.6@640 × 480 px	[61]
Grape	YOLOv4	360 RGB images	88.08	-	[33]
Orange	Faster R-CNN	16,006 RGB images	81.47	85@480 × 270 px	[24]
Passion fruit	Faster R-CNN	2275 RGB-D images	96.2	175@1920 × 1080 px	[23]
Pepper	Mask R-CNN	1127 RGB images	78.16	1180@1280 × 720 px	[62]
Pear	YOLOv4	1337 RGB images.	100	37.8@608 × 608 px	[63]
Strawberry	DeepLabv3+	1000 RGB images	90	-	[64]

2.2.1. Two-Stage Fruit Detection Method

Combining traditional machine learning methods with CNN, the two-stage fruit detection framework is proposed to obtain as many region proposals as possible through a selective search, and use a CNN instead of manually extracting the features of the regions and classifying them. Due to the fine-grained processing of candidate regions, those algorithms usually have high detection accuracy and strong robustness in complex scenarios [65].

To enhance detection capabilities, reduce the missed detection rate of the object and improve the robustness of the model under complex scenes such as the occlusion of the object and the difference in the shape and size of the object, some scholars have improved the two-stage fruit detection algorithm. Tu et al. improved the Faster R-CNN structure by incorporating feature maps from shallower convolution feature maps for regions of interest (ROI) pooling. Its precision and recall of detecting the small object passion fruit with the precision, recall and F1 values were 93.10%, 88.50% and 94.60%, respectively [23]. To improve detection accuracy, Fu et al. improved the Faster R-CNN detection model by two pre-trained network architectures (ZFNet and VGG16) and developed an apple detection algorithm for dense tree canopies. The algorithm structure is shown in Figure 3 [48]. The results displayed that the algorithm based on the VGG16 backbone network performed

better, and its average detection accuracy was 89%. The processing time of each image was 0.181 s, which fully satisfied the requirements of real-time detection by the picking robot.

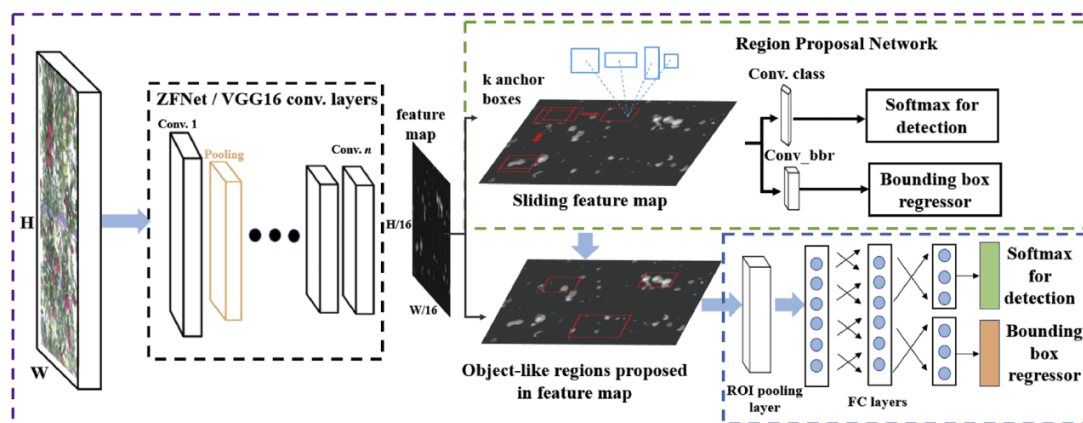


Figure 3. Apple detection algorithm on the Faster R-CNN with the ZFNet/VGG16 [48].

To improve the efficiency and quality of robot harvesting, Divyanth et al. introduced an attention module in the Faster R-CNN model to propose a detection coconut cluster algorithm. The mean average precision (mAP) of the unshaded coconut was 91.2%, and the mAP of the leaf-occluded fruit could reach more than 88.3% [55]. Mai et al. proposed a multiclassifier fusion strategy in conjunction with a Faster R-CNN for small fruit detection. It used features from three different levels to learn three classifiers for object-based classification in the proposal localization in the algorithm (Figure 4). The mAP and F1 values of this model were 77.75% and 81.17%, respectively [24].

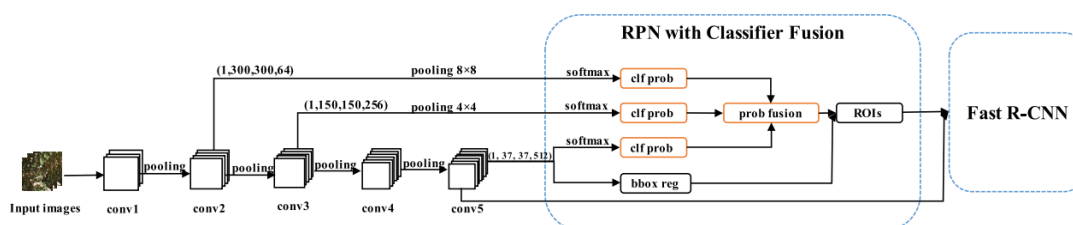


Figure 4. Small fruits detection model diagram based on the Faster R-CNN [24].

Wan et al. proposed a multiclass fruit detection model on the basis of lots of photos of fruit in the orchard, as shown in Figure 5. The model improved the convolution layer and pooling layer of the Faster R-CNN, and the detection speed of each image was 47 ms and the mAP was 86.41% [22].

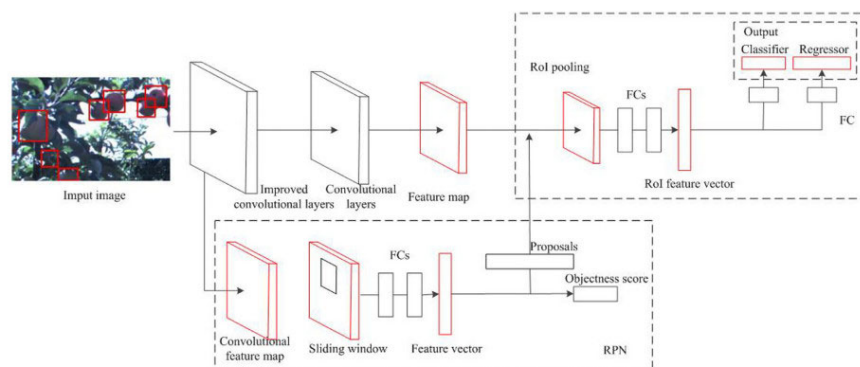


Figure 5. Multiclass fruit detection model [22].

Instance segmentation is the novel approach in object detection that extends the object detection framework by adding an extra branch at the end of the model, enabling instance segmentation for each output suggestion box, allowing segmentation, recognition and localization tasks to be carried out in parallel. As one of common instance segmentation network methods, the mask region convolutional neural network (Mask R-CNN) is composed of Faster R-CNNs and fully convolutional networks (FCNs). During the apple picking process, leaf occlusion and fruit overlap have the most serious impact on the accuracy of apple detection and positioning. The improvement and application of the Mask R-CNN has largely solved the problem of difficulty in recognition caused by fruit occlusion and overlap. To accurately and quickly detect overlapping apples, Jia et al. reported an improved detection model on the basis of Mask R-CNN. The method combined the ResNet and the DenseNet as the backbone network for feature extraction, which could greatly reduce the complexity of the model, and the detection speed was faster with an accuracy of 97.31% [49]. Fruit harvesting needs to be completed at a specific time, otherwise it will affect the quality and storage time of fruits and vegetables. Detection algorithms for agricultural-harvesting robots need to ensure accurate recognition in complex and changing operating environments while ensuring image processing speed. When optimizing the two-stage object detection algorithm, it is necessary to consider how to optimize the model structure and the setting of model parameters while ensuring the accuracy of the model, so that the algorithm cannot only obtain higher detection accuracy but also meet the real-time requirements in real application scenarios [51,61,62,64,66–68].

2.2.2. Single-Stage Detection Method

Although the two-stage object detection algorithm has high detection accuracy, it has a large number of parameters and a large amount of calculation, and the detection speed is slow, making it difficult to complete real-time detection tasks, which affects the working efficiency of the picking robot. In the single-stage detection algorithm, it only uses one CNN to directly predict the classes and locations of different objects for a test image, which greatly reduces the consumption of computing resources and improves the speed of detection, making it very suitable for online real-time detection [69–72].

SSD model is a single-stage detection model based on the VGGNet network and the ResNet network, which has good detection effect and fast speed in object detection and recognition [73,74]. Liang et al. proposed a novel detection algorithm for real-time mango detection. The algorithm was improved on the SSD model of VGG16 and ZFNet for feature extraction network, and the accuracy of the improved algorithm was higher than that of the Faster R-CNN. However, the layer number of the improved network structure was deep and the number of parameters was large, resulting in a slow operation speed [56]. Yuan et al. used inception V2 network as the backbone network and combined it with the SSD model to propose an algorithm for the detection of cherry tomatoes in a greenhouse. The map of this model was 98.85%, which was 6.43% higher than the SSD base model [52]. Agarwal et al. developed an intelligent model for detecting mangoes on trees based on the Darknet-19 backbone network and SSD, as shown in Figure 6. They improved the stability of the work for different lighting conditions by augmenting the input images with data. The mAP of the model was 92.43% in the field [26]. Magalhães et al. compared the detection performance of SSD and YOLO models for tomatoes in greenhouses and found that the SSD model based on the Inception v2 backbone network has better detection performance with an F1 value of 66.15% [53].

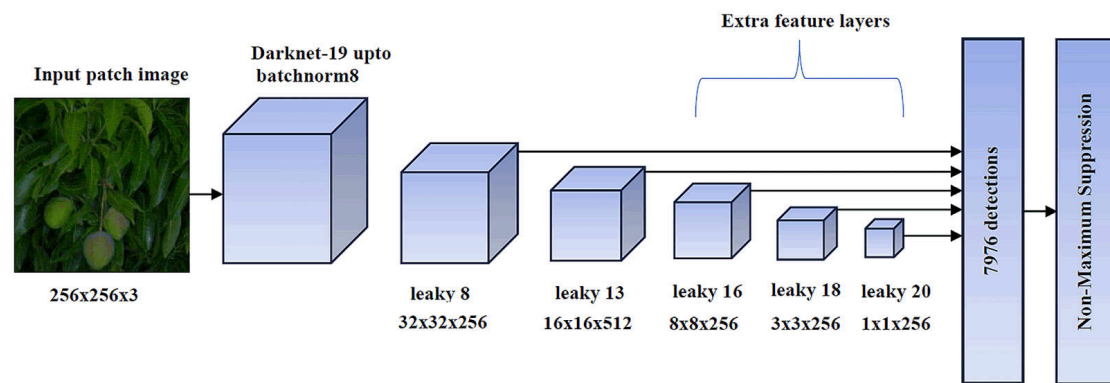


Figure 6. Mango detection model structure based on SSD [26].

Because of the high detection accuracy, short detection time and modular framework design of these algorithms, the YOLO series and its improved algorithms are widely used in various fields such as fruit detection, fruit load estimation, pest detection and disease detection [59,75–80]. The detection principle of the YOLO model is shown in Figure 7. It divides the input image into $S \times S$ grids, and then predicts according to the feature information extracted from the backbone network. Each grid cell predicts multiple bounding boxes along with their confidences and class probabilities. Lai et al. proposed a ripe oil palm fruit detection algorithm based on YOLOV4. Its mAP and recall were 87.9% and 94.3%, respectively [57].

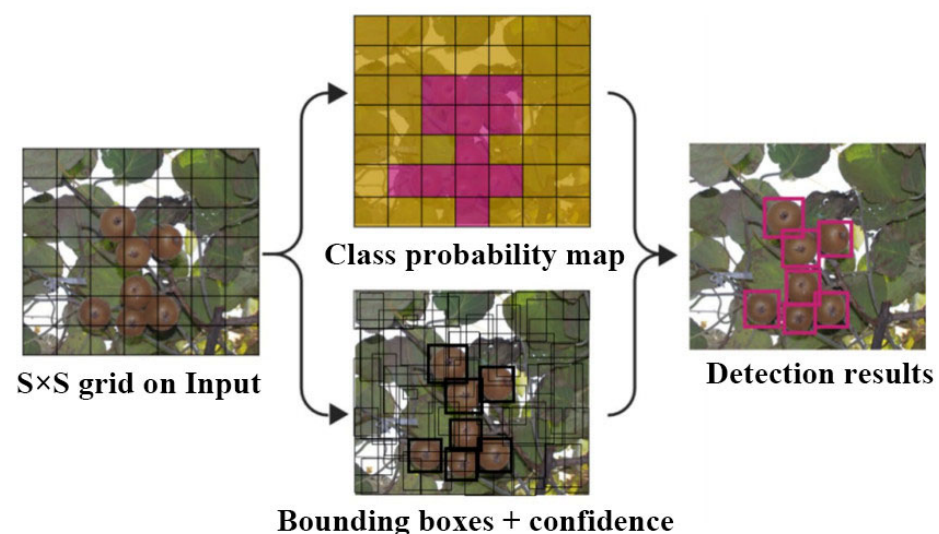


Figure 7. Detection principle of the YOLO model [58].

Lawal replaced the residual linkage module of YOLOv4 and YOLOv5 with a new PANet and FPN to propose an algorithm for real-time detection of greenhouse melons. Harvest experiments displayed that the improved YOLOv4 model has fewer parameters, the detection accuracy was increased by 91.5% and the detection time was reduced to 5 ms [60]. Zhao et al. proposed a lightweight detection model for grape and picking points, as shown in Figure 8. It used the ghost bottleneck instead of the convolutional layer in the residual unit, making this detection algorithm lightweight. Compared with the initial algorithm, the number of parameters was reduced by more than 10%, and the detection error of picking points was reduced by 40 pixels [33].

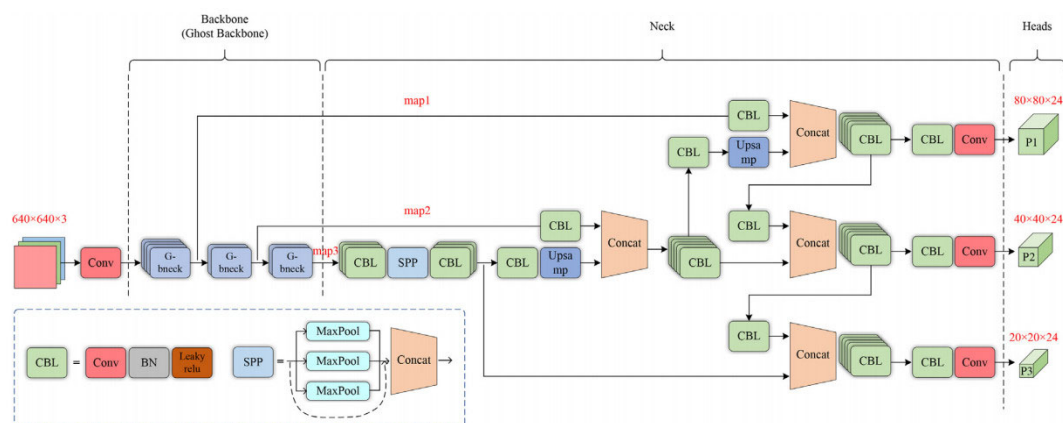


Figure 8. Grape lightweight detection model [33].

The previous deep learning tomato detection algorithm was computationally intensive and highly dependent on high-performance GPUs. Zeng et al. proposed a lightweight detection model to deal with the problem. It used MobileNetV3 as the feature extraction backbone network and improved the down sampling convolution layer, as shown in Figure 9. Field tests showed that the detection speed of this model in the mobile terminal was about 27 frames per second, which was more than 2 times faster than the YOLOv5s model [81]. Some researchers have proposed other fruit detection algorithms such as apple and pear based on the YOLO model [50,54,63,82]. It is worth mentioning that the single-stage detection algorithm is fast in detecting and can meet the real-time requirements of automatic harvesting. However, some old orchards have relatively high fruit tree planting density, the fruit is heavily shaded and the shuttering conditions are poor, which poses a great challenge to fruit detection [83].

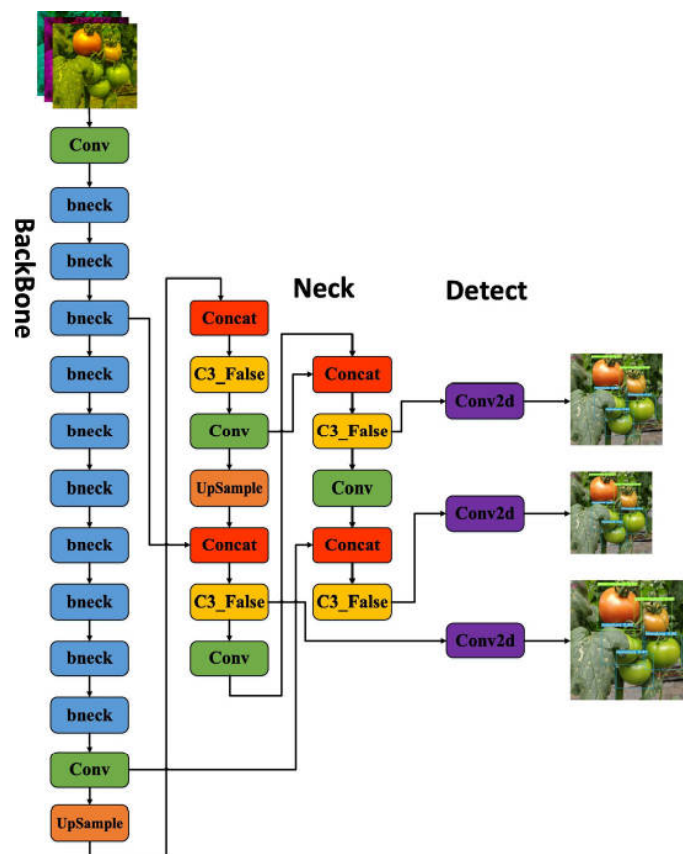


Figure 9. Lightweight tomato detection model [81].

3. Fruit Localization Method

Fruit localization is another important part of the vision system of the fruit- and vegetable-harvesting robot. After the fruit is detected, it is necessary to accurately locate it. In order to help the end-effector realize accurate gripping operation, it is necessary to acquire the data of the picking point of the fruit apple, the three-dimensional position of the fruit and the attitude. In recent years, scholars have proposed many effective localization methods based on stereo vision system, RGB-D camera, etc. in [84–97].

3.1. Fruit Localization Method Based on the Stereo Vision System

It is common to use a binocular or multicamera combination of the stereo vision system to localize the fruit due to the large error of the monocular camera. In the stereo vision system, common stereo matching methods include feature-based methods, region-based methods and so on. Feature-based methods can obtain sparse disparity maps, and dense disparity maps can be obtained by difference estimation, but it is easily affected by fruit occlusion, light, texture repetition, etc., and the difference estimation is computationally large. Region-based methods can obtain dense disparity maps, but it is affected by the affine distortion and radial distortion of the image, and it is difficult to choose the size and shape of the pixel constraint window. Stereo vision matching methods often use binocular cameras, which utilize the principle of triangulation to obtain three-dimensional information about an object through the parallax of two images taken in different orientations [93]. The schematic diagram of binocular stereo vision 3D positioning is shown in Figure 10. Liu et al. recorded the left and right images of pineapple fruit with binocular cameras, and obtained the accurate three-dimensional coordinates of the fruit based on the principle of triangulation. Test results displayed that the absolute positioning error of this method was about 25 mm and the relative positioning error was 1.17% at the distance of 1.7–2.7 m [98].

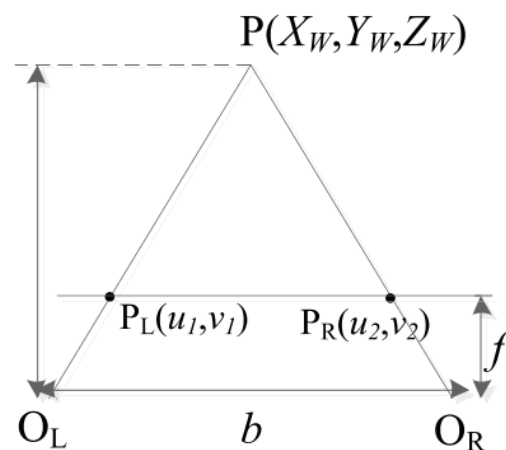


Figure 10. Schematic diagram of binocular stereo vision 3D positioning [98].

In order to accurately locate the fruit and picking points, Luo et al. used two CMOS color cameras to calculate the cluster boundary volume by extracting the 3D spatial coordinates of the detection of the pedicel and the central cut point of the grapes [87]. To address the effects of background and fruit overlap on citrus localization, Hou et al. used Otsu thresholding algorithms and morphological processing methods to restore the full morphology of the fruit. In addition, they obtained the spatial localization information of the fruit based on the Kriging algorithm and combined it with the geometric features of citrus. The results showed that the average error of the localization algorithm was less than 2.75 mm [92]. Considering that multiple cameras can increase the perception range and obtain more image feature information, Chen et al. established four color camera vision

systems, designed a point cloud correction algorithm to calibrate the camera system and developed a point cloud splicing algorithm to improve the accuracy of the reconstructed fruit surface [89].

3.2. Fruit Localization Method Based on RGB-D Cameras

RGB-D cameras can acquire color images and depth images at the same time, and their typical representatives are structured light cameras, time of flight (TOF) cameras, etc. Structured light cameras obtain the position and depth information of an object by emitting an infrared laser of a specific wavelength, projecting it on the object after a certain encoding and calculating the distortion of the returned encoded pattern. TOF cameras emit continuous laser pulses to a target, receive the reflected light with a sensor and obtain the distance to the target based on the round-trip time of flight of the detected light pulses [29]. When the fruit is successfully detected, its 2D information can be obtained. After that, the distance from the camera to the fruit can be obtained from the depth information or the point cloud information to complete the 3D localization of the fruit. Arad et al. used a TOF camera to acquire RGB image information and depth information of bell peppers, and obtained the localization information of the fruits based on the depth information extracted from the detected regions through an associated pixel-world conversion procedure [95]. Li et al. proposed a localization algorithm for occluded apple fruits based on the frustum-based point cloud processing method and YOLACT++ model. The YOLACT++ model was used to locate the fruit in the RGB image. The frustum-based method used the output of the mask and 2D bounding box to estimate the depth of the center of the fruit. Robotic-harvesting results showed that the algorithm was able to reduce the average positioning error by 43% [91]. Bortolotti et al. developed the localization algorithm for apples based on the RGB-D images from the Intel RealSense D435i. The algorithm incorporated the modified YOLOv5 model, the circle detection algorithm and the triangulation method based on depth information. Experimental results displayed that the overall positioning error of the algorithm was 5.7 mm [99]. Lin et al. proposed an algorithm for citrus detection and localization of RGB-D images. The algorithm filtered the image and detected and localized the citrus using the density clustering algorithm and SVM, with an average localization error of about 13 mm in all directions [100]. Accurate positioning of green apples is tricky due to the light, the environment and the fruit color. To address the above problem, Sun et al. proposed a green apple localization algorithm [101]. The gradient field of the target image was analyzed using the image depth information, and the vorticity center and two-dimensional plane projection were built to achieve the localization of the apple center. Then, the apple image was segmented based on the density peak clustering algorithm, and the precise segmentation region of the target image was efficiently obtained using the double sort algorithm. Finally, the above two results were fused to achieve the localization of the target fruit. GAO et al. used the RANSAC algorithm to perform spherical fitting on the point cloud of the fruit surface to obtain the center of the mass of the fruit and improve the accuracy of fruit positioning. The standard deviation of the radius of the sphere fitted by the algorithm for apples was 1.79 mm, and the standard deviation of the center coordinate of the sphere was less than 2.5 mm. The algorithm has an average time of 0.543 s for apple localization, which meets the working requirements of apple picking robots [84]. Liu et al. tested the localization performance of several RGB-D cameras, including the ZED 2i, in a complex kiwifruit orchard. The results showed that the RealSense D435 camera performed the best, with a localization speed of 0.062 s per image and an average error within 19 mm [96].

3.3. Other Fruit Localization Methods

Although there are many ways to obtain image depth information, laser active vision is a better choice considering some characteristics of the perceived image. LiDAR can scan the entire harvest scene and provide relevant ranging information, which has been used by researchers for agricultural automation and robotics. In order to guide the end-effector to grasp the fruit, Tanigaki et al. used an infrared laser ranging sensor to obtain the 3D positions of the target fruit and obstacles and then determine the trajectory of the manipulator's end-effector to avoid collision with obstacles [102]. Considering that RGB cameras are susceptible to factors such as lighting conditions, Genemola et al. used mobile terrestrial laser scanners to detect and locate apples. The positioning accuracy of this method is comparable to that of the RGB system, but it can directly obtain the three-dimensional information of fruit, and is not affected by the change of sunlight [86]. Zhang et al. designed an active positioning system for high-precision fruits, which consisted of an RGB camera, an RGB-D camera, a red line laser, and linear moving parts (Figure 11). It can achieve high precision positioning within 0.6 m–1.2 m and the depth positioning error is less than 4 mm [94]. Experiments have proved that it can locate apples and other fruits more accurately than commercial RealSense cameras [97]. Liu et al. proposed a real-time dynamic fruit detection system based on the ORB-SLAM3 and YOLOv5 algorithms. The ORB-SLAM3 was used to estimate camera attitude in real time. LiDAR was used to measure depth information between the camera and the apple. The results showed that the average positioning error of the system was 21.1 mm [103].

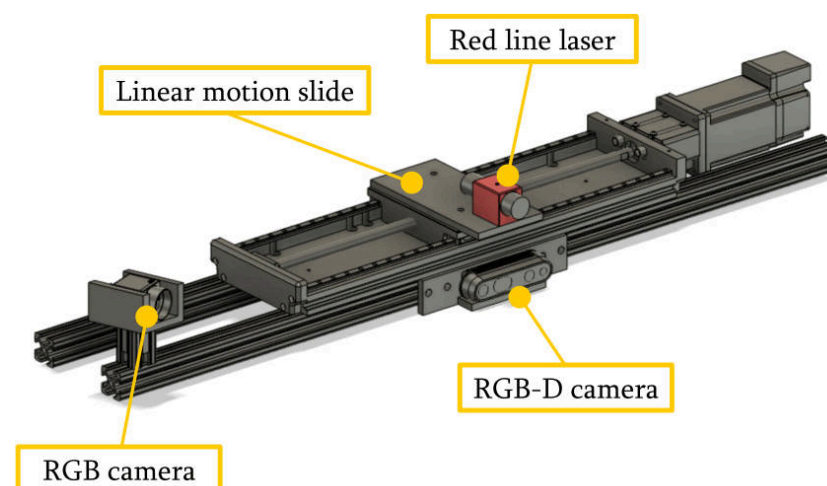


Figure 11. Active positioning system [94].

4. Challenges

The research of fruit recognition and localization based on DL has made great progress, but the existing models still face many problems and challenges before large-scale field application:

- (1) High-quality and large-scale fruit datasets are scarce. Datasets play an irreplaceable role in the development of current object detection algorithms [104]. In order to obtain better and faster detection results, large and comprehensive datasets are indispensable. The number of data samples collected in the fruit detection is required to be large enough for the purpose of excellent detection accuracy. To improve and validate the robustness of the model, the images in the datasets should also contain a variety of scenarios such as single-targeted fruits, multitargeted fruits, branch and stem foliage occlusion, occlusion between fruits and different lighting conditions [105,106]. Fruit growth environments are mostly complex and unstructured and fruit growth is

cyclical, and data collection must be completed within a specific time frame, which brings great challenges to the establishment of large-scale datasets [39,70,107].

- (2) The stability, generalization and robustness of detection and localization algorithms in complex scenarios need to be further improved. Because the area occupied by small fruits in the image is small, the features that can be extracted after the convolution operation are limited and easily lost, and some algorithms perform poorly in dense and occluded scenes. Due to the existence of various complex and variable scenarios in the natural environment where the automated-harvesting robot operates, the model used for fruit detection and localization needs to have both high stability, generalizability and robustness to ensure its stable operational effect and high operational efficiency [108–114]. Therefore, improving the performance of models in complex scenarios while ensuring the detection and localization accuracy is currently a challenge in the field of object recognition.
- (3) There is a high degree of complexity and uncertainty in the agricultural environment. Although CNNs based on DL can solve the object detection and recognition problem well, the neural network has a huge demand for the training dataset, and the model training is computationally intensive and time-consuming. In addition, although the correct rate of object recognition is relatively high, due to the complexity and uncertainty of the agricultural environment, the convolutional neural network still has a large error in object localization [85,115]. As the growth state of fruits and vegetables is affected by the environment and changes in the characteristics of the fruits and vegetables themselves and other factors, the detection and localization algorithms need to take into account more factors and the corresponding CNN structure will be more complex, which will result in longer model processing times, poorer real-time performance and lower operational efficiency of the robot [32,33]. There are many external disturbances and obstructions in the fruit-picking and -harvesting environment, which can cause large random errors to the machine vision localization system, and the pattern of these errors is difficult to obtain and compensate.

5. Future Trends

- (1) Research and development of a weakly supervised model. Due to the time-intensive and labor-intensive nature of the data collection and labeling tasks, and the difficulties associated with large-scale data collection tasks, utilizing fewer sample data for model training and reducing the amount of labeled data is important for reducing labor costs and increasing the flexibility of detection. For some fruit datasets that are difficult to collect and small in number, their intrinsic connection with common large-scale datasets can be utilized for migration learning. Relatively few models for small-scale datasets have been conducted on the semi-supervised and unsupervised models for fruit detection. The data-labeling process based on supervised learning is time-consuming, labor-intensive and inefficient. In addition, for certain specific recognition tasks, the data-labeling process requires guidance from relevant experts. Using small- and medium-scale data for semi-supervised learning can obtain good detection results. Therefore, weakly supervised models will be the future development direction of fruit recognition models [39,116–118].
- (2) Development of lightweight detection and localization model for automated fruit- and vegetable-harvesting robot. Researchers are actively exploring lightweight technologies such as compression algorithms, quantization methods and depth-separable convolution to reduce the computational complexity and storage space occupied by the model, so that object detection technology can better adapt to the needs of actual scenarios. Under the premise of accuracy, models for embedded devices not only

- require a small model size, but also ensure speed to achieve real-time detection and localization [119]. Therefore, the subsequent research should focus on improving the detection and localization speed based on lightweight models such as MobileNet, ESPNet and GhostNet on embedded devices with limited computing power [120–123].
- (3) Fruit detection and localization based on multiple sensor fusion. Stereo vision devices provide high detection accuracy, but they are susceptible to light conditions in outdoor environments and the vision-matching process is complex and takes a long time to compute. LiDARs have excellent localization accuracy and high reliability compared to other sensors, but have the disadvantages of slow speed and high cost. The use of multisensor information fusion technology can greatly improve the detection accuracy and reduce the localization error [6]. By extracting information from multiple sensors, the information of the automated fruit- and vegetable-harvesting robot can be more comprehensively obtained, and combined with the optimal object recognition and localization algorithm, the recognition and localization accuracy can be effectively improved with high reliability and robustness. In recent years, new visual sensors such as TOF cameras, light field cameras and chlorophyll fluorescence cameras have attracted more and more researchers' attention. Researchers can expand the image dataset of fruit and vegetable images or extract fruit signature features and make target region judgment, then make decision fusion according to the influence of occlusion or lighting conditions and finally merge to complete object detection and positioning, which can effectively reduce the demand for data and the calculation amount of the neural network, and shorten the network training time [22].

6. Conclusions

Accurate identification and precise localization of fruits is crucial for fruit-harvesting robots. In the past, the traditional fruit detection technology had poor accuracy and a low detection speed, which cannot meet the needs of harvesting robots for real-time fruit detection and localization. Deep learning technology can solve the above problems and play a unique role in automated agricultural harvesting. The methods in data collection, data enhancement and data annotation for the dataset production are summarized and data enhancement techniques based on DL are widely used. Currently, the stereo vision system and RGB-D cameras are often used to localize the position of fruits, which can be used in indoor or outdoor scenes and are less affected by changes in lighting conditions and object features, while meeting the demand for a higher localization accuracy of picking robots. It is shown in studies that light variations, variable crop growth states, shading and clustering in natural environments are the main challenges for accurate recognition and localization. Currently, deep learning algorithm training often requires a large number of samples. In the future, researchers need to propose algorithms with higher accuracy, a faster processing speed and lighter weight based on geometric features, image features, etc. In addition to this, deep learning algorithms that deal with small datasets and weakly supervised algorithms need to be further developed. Due to the complexity and uncertainty of agricultural environments, harvesting robot vision systems still have large localization errors. This requires the utilization of multisensor fusion technology and artificial intelligence technology to improve the integrated visual error tolerance and anti-interference capability.

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